

M R MADHAVAN

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EDUCATION

VIT University

Bachelor of Technology

Major in Computer Science; Minors in Artificial Intelligence and Robotics

GPA: 8.48/10.0

Relevant Coursework: Software Engineering; Operating Systems; Algorithms; Artificial Intelligence; Data Structures

Chennai, IN

June 2025

PROFESSIONAL EXPERIENCE

Shiash Info Solutions llc

AI Engineer Intern

Chennai, IN

Aug 2023 – Jun 2024

- Implemented a GenAI image classification system using python for 25,000+ images with accurate training pipeline; improved accuracy and reduced wrong predictions; used OpenCV, Tensorflow for image detection.
- Overhauled data processing and training of models, resulting in increased usability and reduced training run time by 25%; supported RAG (Retrieval-Augmented Generation) pipeline using data ingestion.
- Analyzed data from 25,000 monthly active users and used outputs; increased average app engagement time by 2x, 30% decrease in drop off rate; used LLM, fine-tuning, embeddings, prompt engineering.
- Initiated comprehensive suite of unit tests for a critical software module, achieving 100% test coverage.

Capabl India llc

Software Development Engineer Intern

Bangalore, IN

Jun 2023 – Jul 2023

- Developed a library management system using python and Azure, automating 90% of manual work and streamlining operations using Docker, Kubernetes, CI/CD and updated Sql databases accordingly.
- Refactored data pipelines and training scripts, cutting model training time by 25% and increasing overall usability and designed a mobile application using TypeScript, Android Studio, React Native.
- Designed a web application using Angular and REST APIs using Flask implemented 50+ E2E tests using Selenium, simulating each user group's actions.
- Executed frontend testing processes that cut UI-related bugs by 15%.

PATENTS

“SYSTEM TO SECURE MONEYTRANSACTION” - (published) Indian Patent Application No. 202541034231

- Created fingerprint and OTP-based multi-factor authentication for UPI transactions, mitigating 24% of simulated phishing attacks by using secure API.
- Conducted 100+ test scenarios to confirm transaction reliability and regulatory adherence.

“REAL-TIME ATM FRAUD DETECTION” - (published) Indian Patent Application No. 202541109062

- Deployed a computer vision system to detect ATM fraud using mask detection and gesture analysis.
- Applied real-time video analysis to identify threats, reducing false positives by 22%.

UNIVERSITY PROJECTS

ByteBasket Online Grocery Store

May 2025

- Engineered an e-commerce platform with PHP, MySQL, and Bootstrap, improving order experience by 40%.
- Launched product search, filtering, and real-time cart updates, reducing drop-off rate by 25%.
- Enhanced non-technical access to data by 40% and lowered analyst workload.

Natural Language SQL Chatbot

Aug 2024

- Built a chatbot that converted plain English into SQL queries with over 95% accuracy.
- Enabled support for 50+ query types using intent classification and slot filling.
- Validated system integrity with 100+ test scenarios for compliance.

SKILLS

- **Technical** : Python, Java, JavaScript, RDBMS, Angular, HTML5, CSS, React.js, GenAI, REST API, FastAPI, Flask, TypeScript, React Native, Android Studio
- **Tools & Platforms**: AWS, Azure, Docker, Kubernetes, Git & GitHub, Jenkins, Postman, Visual Studio Code, PostgreSQL, Oracle, Spring, JSF, JUnit, Mockito, TestNG, Maven, Gradle, Hive, Starburst, Hadoop, Kafka